

Fundamentals of Electric Circuits

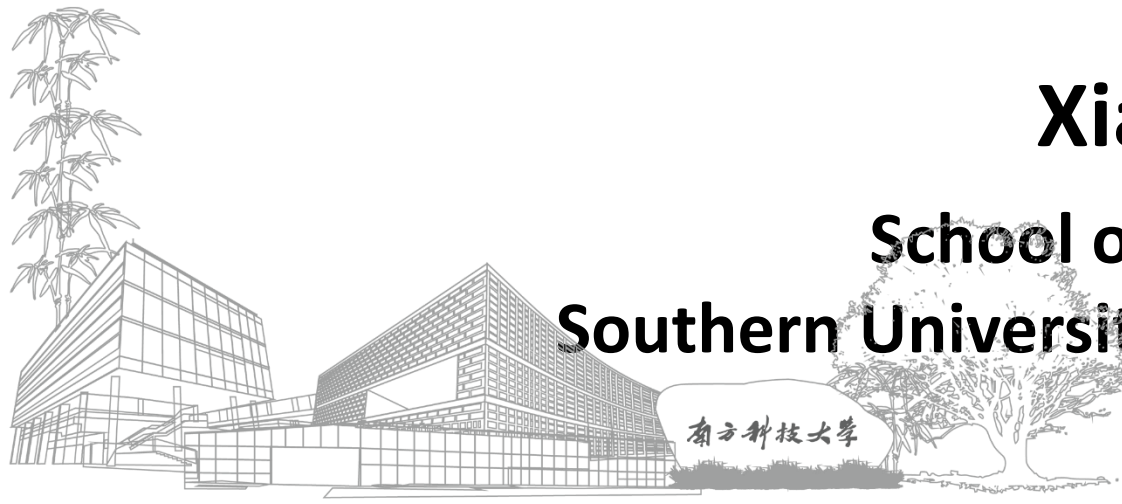
电路基础

Lecture9: Sinusoids and Phasors

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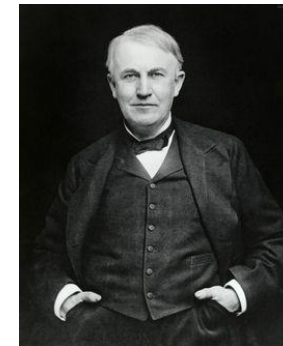


Outline

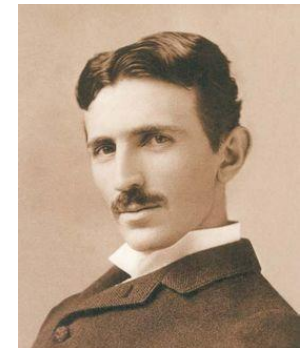
- **Alternating Current**
- **Sinusoids**
- **Phasors**
- **Phasor Relationships for Circuits Elements**
- **Impedance and Admittance**

Introduction

- Thus far our analysis has been limited for the most part to DC circuits: those circuits excited by constant or time-invariant sources. We now begin the analysis of circuits in which the source voltage or current is time-varying
- **Alternating Current (AC, ac, 交流电):** AC is a **sinusoidal current (正弦交流电)**, meaning the **current reverses at regular times (电流方向在交替变化)** and has **alternating positive and negative values (电流有正有负)**
- In the late 1800's there was a battle between proponents of DC and AC. AC won out due to its efficiency for long distance transmission
- Nowadays, alternating current has been the dominant form of electrical power that is delivered to homes and industry



Thomas Alva Edison



Nikola Tesla

Sinusoids (I)

- **Sinusoids (正弦量)** are interesting to us because there are a number of natural phenomenon that are sinusoidal in nature. Also, through Fourier analysis, any practical periodic function can be made by adding sinusoids
- A sinusoidal voltage can be represented as

$$v(t) = V_m \sin \omega t$$

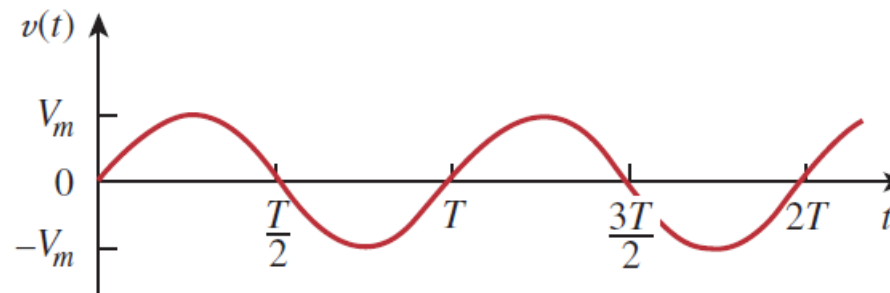
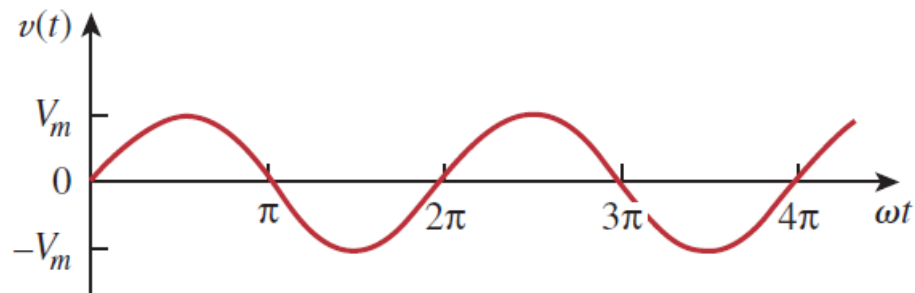
where

振幅、最大值 V_m = the *amplitude* of the sinusoid

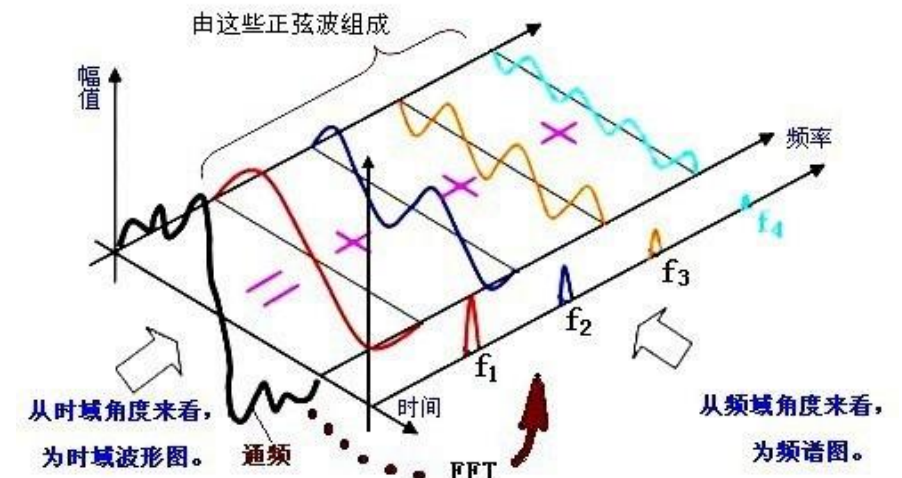
角频率 ω = the *angular frequency* in radians/s

瞬时幅角、相位 ωt = the *argument* of the sinusoid

$$T = \frac{2\pi}{\omega}$$



$$v(t + T) = v(t)$$



Sinusoids (II)

- A sinusoidal voltage can be represented as

$$v(t) = V_m \sin \omega t$$

- Period: $T = \frac{2\pi}{\omega}$

- The period is inversely related to another important characteristic, the frequency:

$$f = \frac{1}{T}$$

- The units of this is **cycles per second**, or **Hertz (Hz)**
- It is often useful to refer to frequency in angular terms

$$\omega = 2\pi f$$



Heinrich Rudolf Hertz
(1857-1894)
German Experimental
Physicist

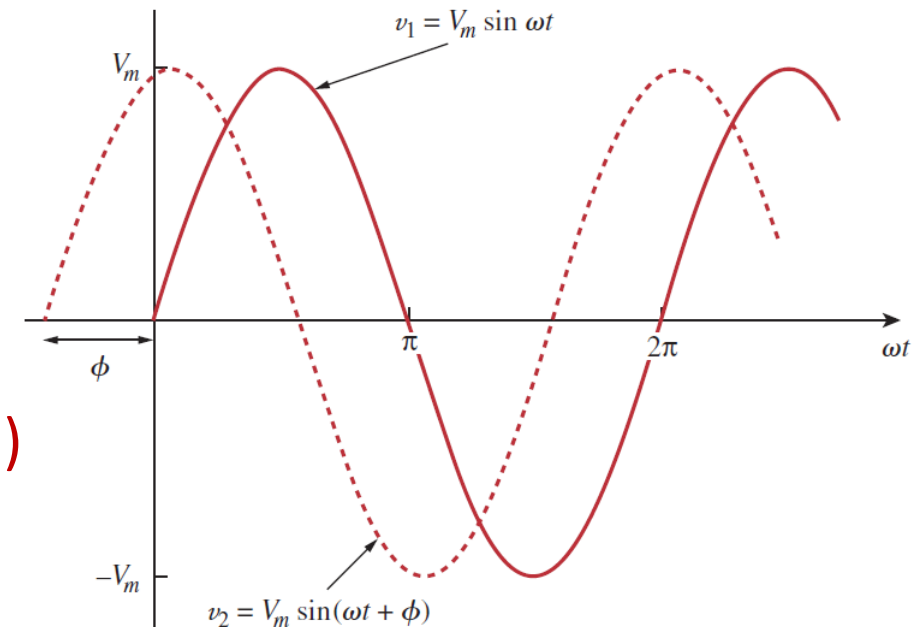
Sinusoids (III)

- Let's consider a more general expression for the sinusoid

$$v(t) = V_m \sin(\omega t + \phi)$$

where $(\omega t + \phi)$ is the argument and ϕ is the *phase*. 初相位

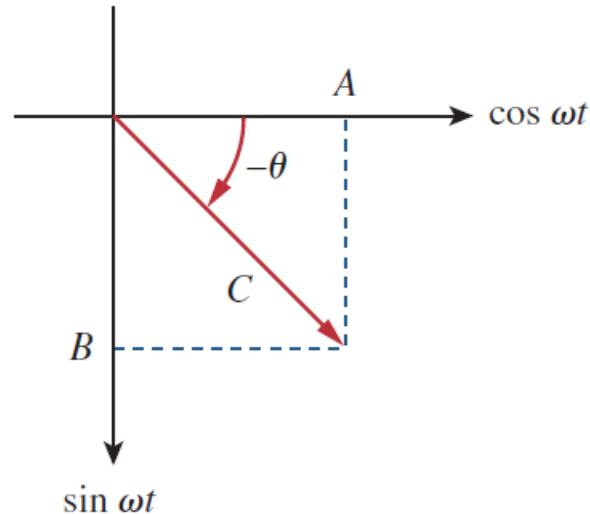
- We say that v_2 **leads** v_1 by ϕ or that v_1 **lags** v_2 by ϕ
- If $\phi \neq 0$, we also say that v_1 and v_2 are **out of phase (异相)**. If $\phi = \pm 180^\circ$, **antiphase (反相)**
- If $\phi = 0$, then v_1 and v_2 are said to be **in phase (同相)**
- If two sinusoids are in phase, then this means that they reach their maximum and minimum at the same time



Sinusoids (IV)

- Sinusoids may be expressed as sine or cosine. The conversion between them is

$$\begin{aligned}
 \sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\
 \cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B
 \end{aligned}
 \Rightarrow
 \begin{aligned}
 \sin(\omega t \pm 180^\circ) &= -\sin \omega t \\
 \cos(\omega t \pm 180^\circ) &= -\cos \omega t \\
 \sin(\omega t \pm 90^\circ) &= \pm \cos \omega t \\
 \cos(\omega t \pm 90^\circ) &= \mp \sin \omega t
 \end{aligned}$$



$$A \cos \omega t + B \sin \omega t = C \cos(\omega t - \theta)$$

$$C = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1} \frac{B}{A}$$

Example: Sinusoids

Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

Example 9.2

■ **METHOD 1** In order to compare v_1 and v_2 , we must express them in the same form. If we express them in cosine form with positive amplitudes,

$$\begin{aligned} v_1 &= -10 \cos(\omega t + 50^\circ) = 10 \cos(\omega t + 50^\circ - 180^\circ) \\ v_1 &= 10 \cos(\omega t - 130^\circ) \quad \text{or} \quad v_1 = 10 \cos(\omega t + 230^\circ) \end{aligned} \quad (9.2.1)$$

and

$$\begin{aligned} v_2 &= 12 \sin(\omega t - 10^\circ) = 12 \cos(\omega t - 10^\circ - 90^\circ) \\ v_2 &= 12 \cos(\omega t - 100^\circ) \end{aligned} \quad (9.2.2)$$

It can be deduced from Eqs. (9.2.1) and (9.2.2) that the phase difference between v_1 and v_2 is 30° . We can write v_2 as

$$v_2 = 12 \cos(\omega t - 130^\circ + 30^\circ) \quad \text{or} \quad v_2 = 12 \cos(\omega t + 260^\circ) \quad (9.2.3)$$

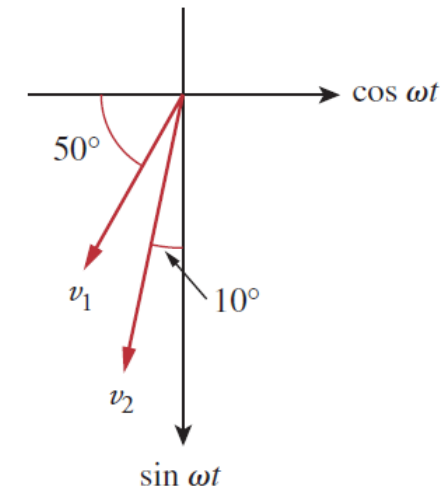
Comparing Eqs. (9.2.1) and (9.2.3) shows clearly that v_2 leads v_1 by 30° .

■ **METHOD 2** Alternatively, we may express v_1 in sine form:

$$\begin{aligned} v_1 &= -10 \cos(\omega t + 50^\circ) = 10 \sin(\omega t + 50^\circ - 90^\circ) \\ &= 10 \sin(\omega t - 40^\circ) = 10 \sin(\omega t - 10^\circ - 30^\circ) \end{aligned}$$

But $v_2 = 12 \sin(\omega t - 10^\circ)$. Comparing the two shows that v_1 lags v_2 by 30° . This is the same as saying that v_2 leads v_1 by 30° .

■ **METHOD 3** We may regard v_1 as simply $-10 \cos \omega t$ with a phase shift of $+50^\circ$. Hence, v_1 is as shown in Fig. 9.5. Similarly, v_2 is $12 \sin \omega t$ with a phase shift of -10° , as shown in Fig. 9.5. It is easy to see from Fig. 9.5 that v_2 leads v_1 by 30° , that is, $90^\circ - 50^\circ - 10^\circ$.



Outline

- Alternating Current
- Sinusoids
- **Phasors**
- **Phasor Relationships for Circuits Elements**
- Impedance and Admittance

Phasors (I)

- **Phasors (phase vector, 相量):** a phasor is a complex number that represents the amplitude and phase of a sinusoid
- **Complex number (复数):** a complex number z can be represented in rectangular form as

$$z = x + jy$$

➤ where $j = \sqrt{-1}$ being the imaginary unit; x and y the real and the imaginary part of z

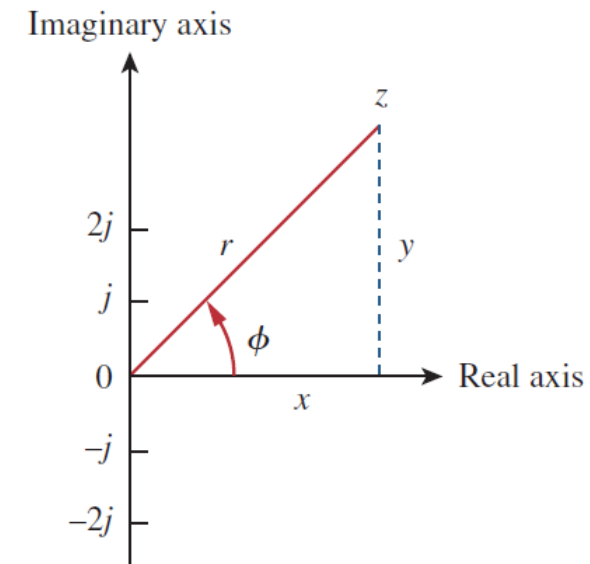
$$z = x + jy \quad \text{Rectangular form} \quad \text{直角坐标}$$

$$z = r \angle \phi \quad \text{Polar form} \quad \text{极坐标}$$

$$z = re^{j\phi} \quad \text{Exponential form} \quad \text{指数形式}$$

Modulus (模) $r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x} \quad \text{Argument (幅角)}$

$$z = x + jy = r \angle \phi = r(\cos \phi + j \sin \phi)$$



Phasors (II)

- The following mathematical operations hold for complex numbers

Addition: $z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$

Subtraction: $z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$

Multiplication: $z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2$

Division: $\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$

Reciprocal: $\frac{1}{z} = \frac{1}{r} \angle -\phi \quad \Rightarrow \quad \frac{1}{j} = -j$

Square root: $\sqrt{z} = \sqrt{r} \angle \phi/2$

Complex conjugate: $z^* = x - jy = r \angle -\phi = r e^{-j\phi}$

Phasors (III)

- From the Euler formula, we have

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi \Rightarrow \begin{aligned} \cos \phi &= \operatorname{Re}(e^{j\phi}) \\ \sin \phi &= \operatorname{Im}(e^{j\phi}) \end{aligned}$$

- Thus a sinusoid can be written as

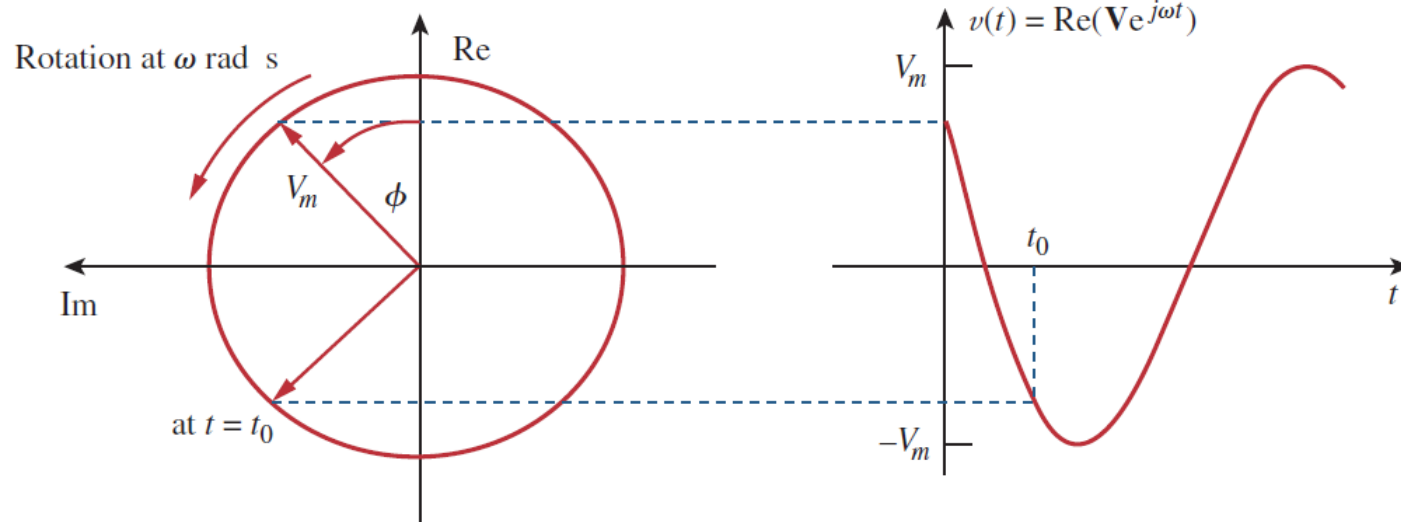
$$v(t) = V_m \cos(\omega t + \phi) = \operatorname{Re}(V_m e^{j(\omega t + \phi)})$$

$$v(t) = \operatorname{Re}(V_m e^{j\phi} e^{j\omega t})$$



$$v(t) = \operatorname{Re}(\mathbf{V} e^{j\omega t})$$

$$\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi$$



$$v(t) = V_m \cos(\omega t + \phi) \Leftrightarrow \begin{aligned} \mathbf{V} &= V_m \angle \phi \\ \text{(Time-domain representation)} &\quad \quad \quad \text{(Phasor-domain representation)} \end{aligned}$$

Sinusoid-Phasor Transformation (I)

- Here is a handy table for transforming various time domain sinusoids into phasor domain

TABLE 9.1

Sinusoid-phasor transformation.

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$

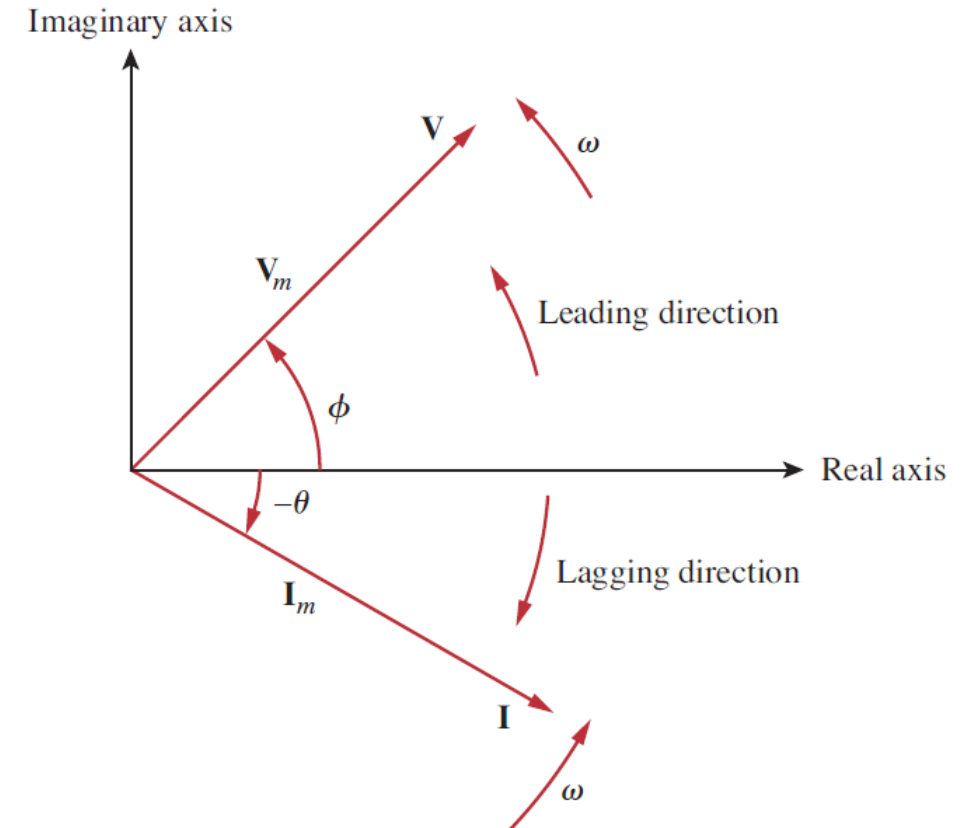


Figure 9.8

A phasor diagram showing $\mathbf{V} = V_m \angle \phi$ and $\mathbf{I} = I_m \angle -\theta$.

Sinusoid-Phasor Transformation (II)

- Differentiating and integrating

$$v(t) = \text{Re}(\mathbf{V}e^{j\omega t}) = V_m \cos(\omega t + \phi)$$

$$\begin{aligned}\frac{dv}{dt} &= -\omega V_m \sin(\omega t + \phi) = \omega V_m \cos(\omega t + \phi + 90^\circ) \\ &= \text{Re}(\omega V_m e^{j\omega t} e^{j\phi} e^{j90^\circ}) = \text{Re}(j\omega \mathbf{V} e^{j\omega t})\end{aligned}$$

$$\begin{array}{ccc}\frac{dv}{dt} & \Leftrightarrow & j\omega \mathbf{V} \\ \text{(Time domain)} & & \text{(Phasor domain)}\end{array}$$

- Differentiating a sinusoid is equivalent to multiplying its corresponding phasor by $j\omega$

$$\begin{array}{ccc}\int v dt & \Leftrightarrow & \frac{\mathbf{V}}{j\omega} \\ \text{(Time domain)} & & \text{(Phasor domain)}\end{array}$$

- Integrating a sinusoid is equivalent to dividing its corresponding phasor by $j\omega$

Phasor Relationships for Resistors

- For the resistor, the voltage and current are related via Ohm's law
- As such, the voltage and current are in phase with each other

$$v = iR = RI_m \cos(\omega t + \phi)$$

- The phasor form of this voltage is

$$\mathbf{V} = RI_m \angle \phi$$

- The phasor representation of the current is $\mathbf{I} = I_m \angle \phi$
- Hence

$$\mathbf{V} = R\mathbf{I}$$

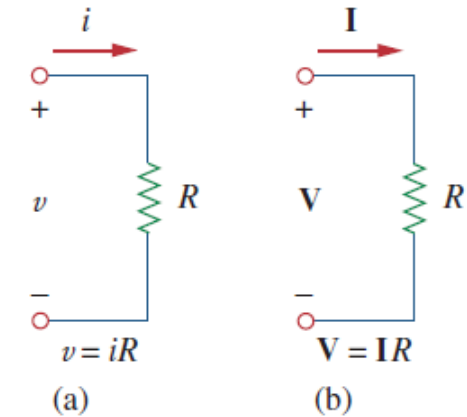
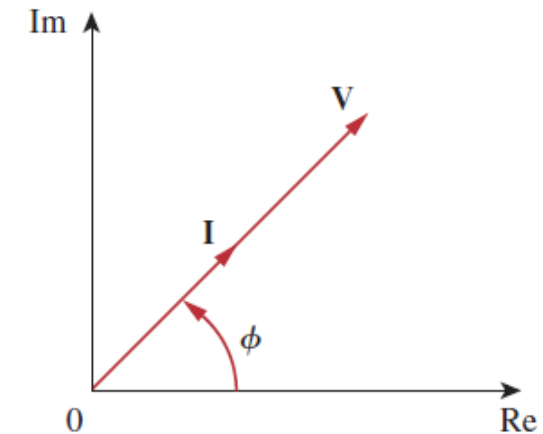


Figure 9.9

Voltage-current relations for a resistor in the: (a) time domain, (b) frequency domain.

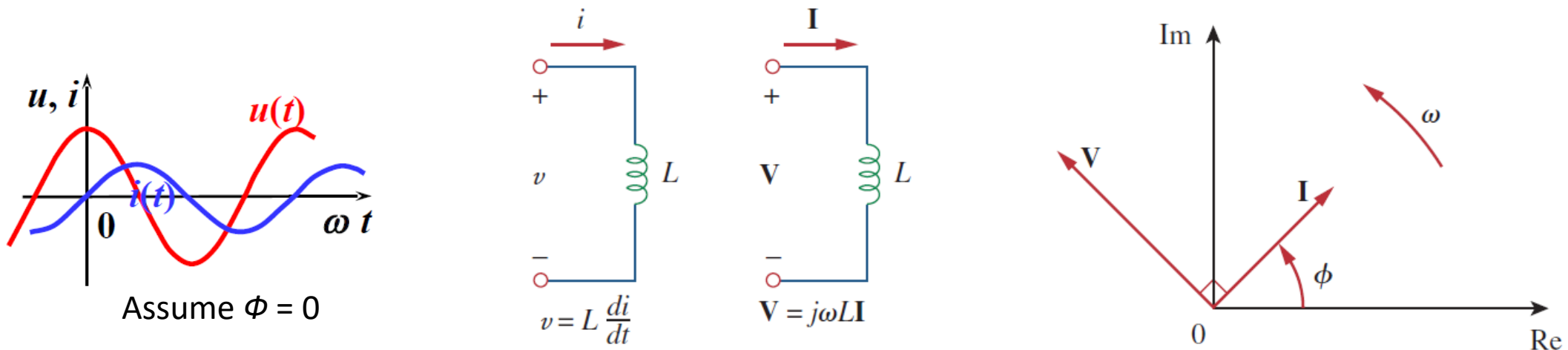


Phasor Relationships for Inductors

- Inductors on the other hand have a phase shift between the voltage and current
- In this case, **the voltage leads the current by 90 degree**. Or one says the current lags the voltage, which is the standard convention
- This is represented on the phasor diagram by a positive phase angle between the voltage and current

$$v = L \frac{di}{dt} = -\omega L I_m \sin(\omega t + \phi) = \omega L I_m \cos(\omega t + \phi + 90^\circ)$$

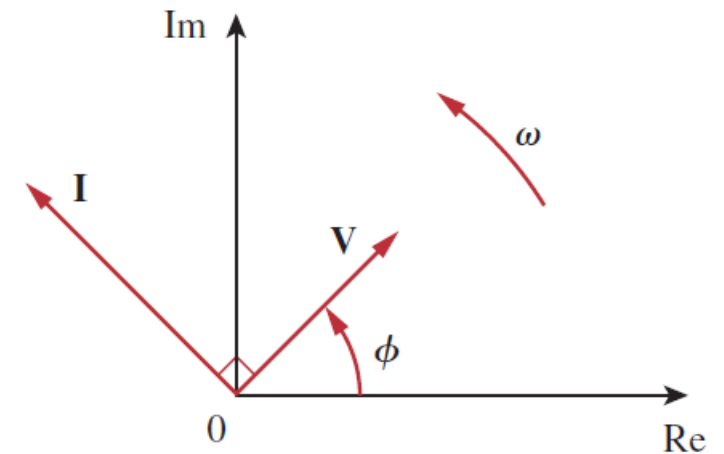
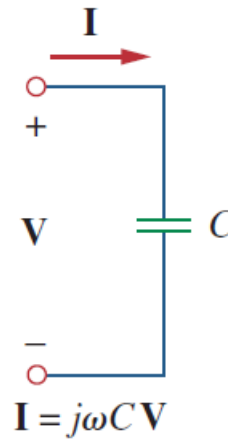
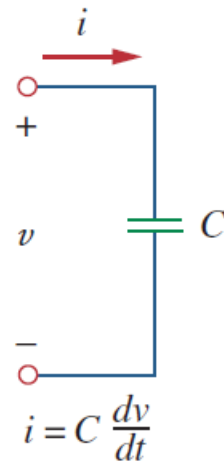
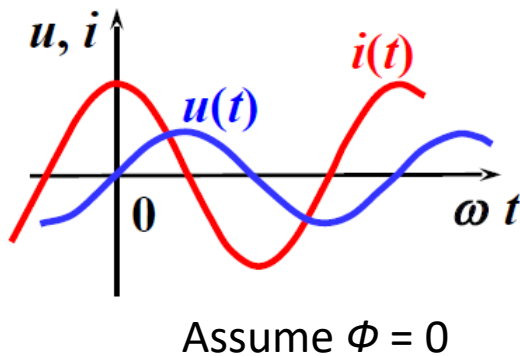
$$\mathbf{V} = \omega L I_m e^{j(\phi + 90^\circ)} = \omega L I_m e^{j\phi} e^{j90^\circ} = \omega L I_m \angle \phi + 90^\circ \Rightarrow \mathbf{V} = j\omega L \mathbf{I}$$



Phasor Relationships for Capacitors

- Capacitors have the opposite phase relationship as compared to inductors
- In this case, the current leads the voltage
- In a phasor diagram, this corresponds to a negative phase angle between the voltage and current

$$i = C \frac{dv}{dt} \quad \mathbf{I} = j\omega C \mathbf{V} \quad \Rightarrow \quad \mathbf{V} = \frac{\mathbf{I}}{j\omega C}$$



Summary: Voltage current relationships

TABLE 9.2

Summary of voltage-current relationships.

Element	Time domain	Frequency domain
R	$v = Ri$	$\mathbf{V} = R\mathbf{I}$
L	$v = L\frac{di}{dt}$	$\mathbf{V} = j\omega L\mathbf{I}$
C	$i = C\frac{dv}{dt}$	$\mathbf{V} = \frac{\mathbf{I}}{j\omega C}$

Outline

- Alternating Current
- Sinusoids
- Phasors
- Phasor Relationships for Circuits Elements
- **Impedance and Admittance**

Generalized Ohm's Law

Resistor

$$\mathbf{V} = R\mathbf{I},$$

$$\frac{\mathbf{V}}{\mathbf{I}} = R,$$

Inductor

$$\mathbf{V} = j\omega L\mathbf{I},$$

$$\frac{\mathbf{V}}{\mathbf{I}} = j\omega L,$$

Capacitor

$$\mathbf{V} = \frac{\mathbf{I}}{j\omega C}$$

$$\frac{\mathbf{V}}{\mathbf{I}} = \frac{1}{j\omega C}$$

- From these three expressions, we obtain Ohm's law in phasor form for any type of elements as

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} \quad \text{or} \quad \mathbf{V} = \mathbf{Z}\mathbf{I}$$

- Where $\mathbf{Z}$ is a frequency-dependent quantity known as *impedance*, measure in ohms
- Impedance 阻抗 and Admittance 导纳

Impedance & Admittance

- Impedance (阻抗) is a complex number that can be written as

$$\mathbf{Z} = R + jX$$

- Where $R = \text{Re}(\mathbf{Z})$ is the resistance (电阻) and $X = \text{Im}(\mathbf{Z})$ is the reactance (电抗)
- We say the impedance is inductive when the reactance X is positive or capacitive when X is negative
- The impedance, resistance, and reactance are all measured in ohms. Being a complex number, the impedance may also be expressed in polar form as

$$\mathbf{Z} = R + jX = |\mathbf{Z}| \angle \theta$$

$$|\mathbf{Z}| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1} \frac{X}{R}$$

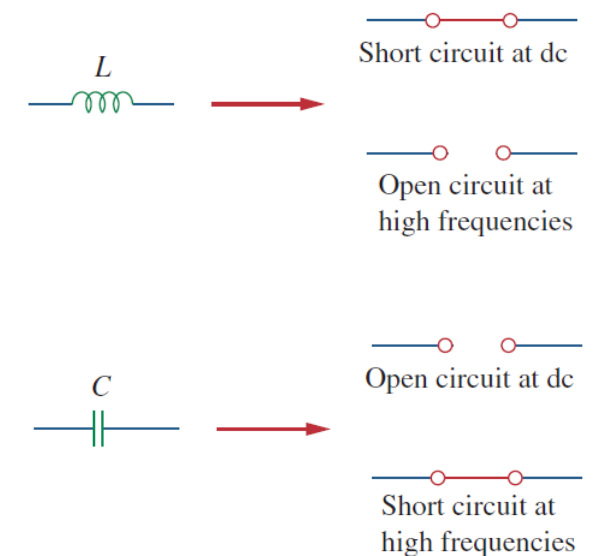
- Admittance (导纳) is simply the inverse of impedance

- Admittance = Conductance (电导) + Susceptance (电纳)

TABLE 9.3

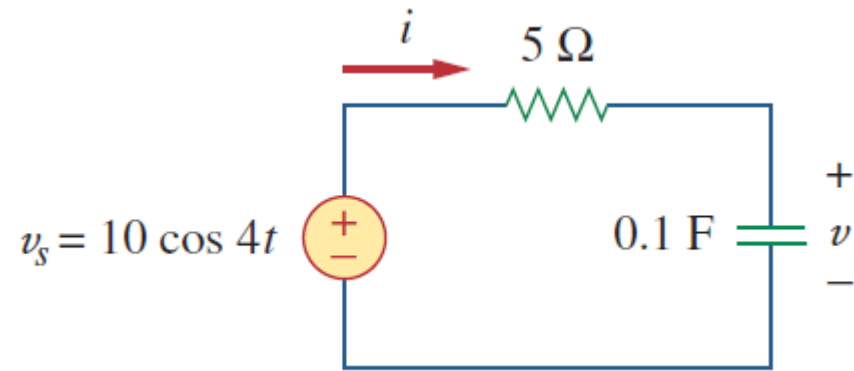
Impedances and admittances of passive elements.

Element	Impedance	Admittance
R	$\mathbf{Z} = R$	$\mathbf{Y} = \frac{1}{R}$
L	$\mathbf{Z} = j\omega L$	$\mathbf{Y} = \frac{1}{j\omega L}$
C	$\mathbf{Z} = \frac{1}{j\omega C}$	$\mathbf{Y} = j\omega C$



Example: Generalized Ohm's Law

- Q: Find $v(t)$ and $i(t)$ in the circuit



Solution:

From the voltage source $10 \cos 4t$, $\omega = 4$,

$$\mathbf{V}_s = 10 \angle 0^\circ \text{ V}$$

The impedance is

$$\mathbf{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5 \Omega$$

Hence the current

$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{10 \angle 0^\circ}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2} \\ &= 1.6 + j0.8 = 1.789 \angle 26.57^\circ \text{ A} \end{aligned}$$

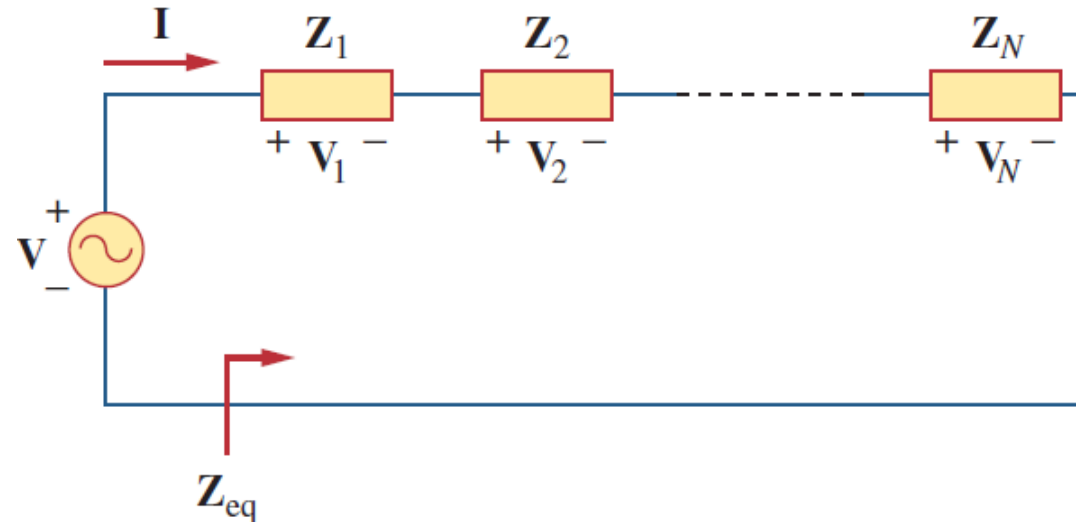
The voltage across the capacitor is

$$\begin{aligned} \mathbf{V} &= \mathbf{I} \mathbf{Z}_C = \frac{\mathbf{I}}{j\omega C} = \frac{1.789 \angle 26.57^\circ}{j4 \times 0.1} \\ &= \frac{1.789 \angle 26.57^\circ}{0.4 \angle 90^\circ} = 4.47 \angle -63.43^\circ \text{ V} \end{aligned}$$

$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

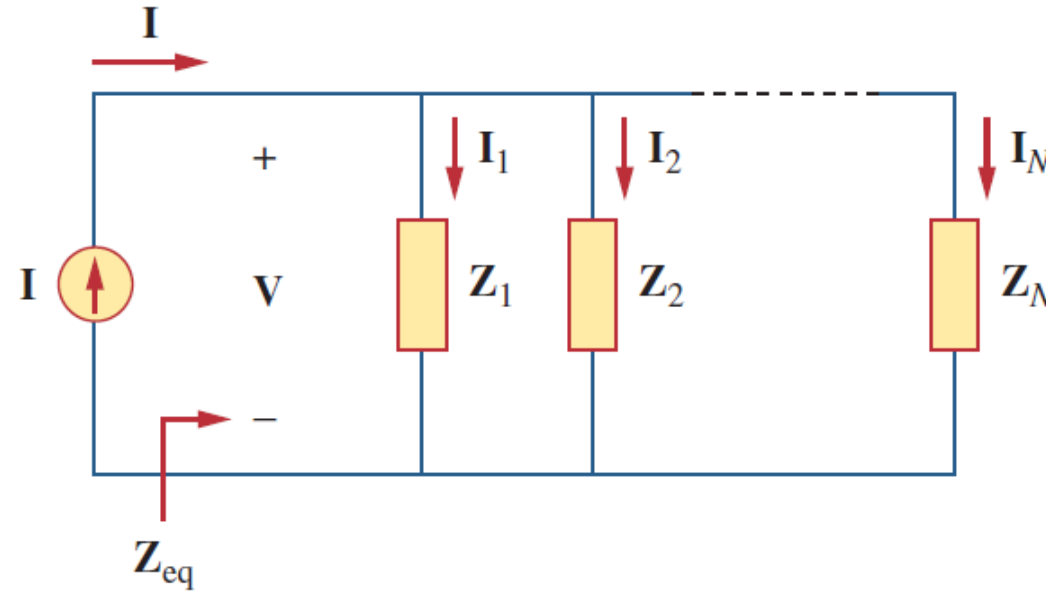
Impedance of Series Circuit



$$V = V_1 + V_2 + \cdots + V_N = I(Z_1 + Z_2 + \cdots + Z_N)$$

$$Z_{eq} = \frac{V}{I} = Z_1 + Z_2 + \cdots + Z_N$$

Impedance & Admittance of Parallel Circuits



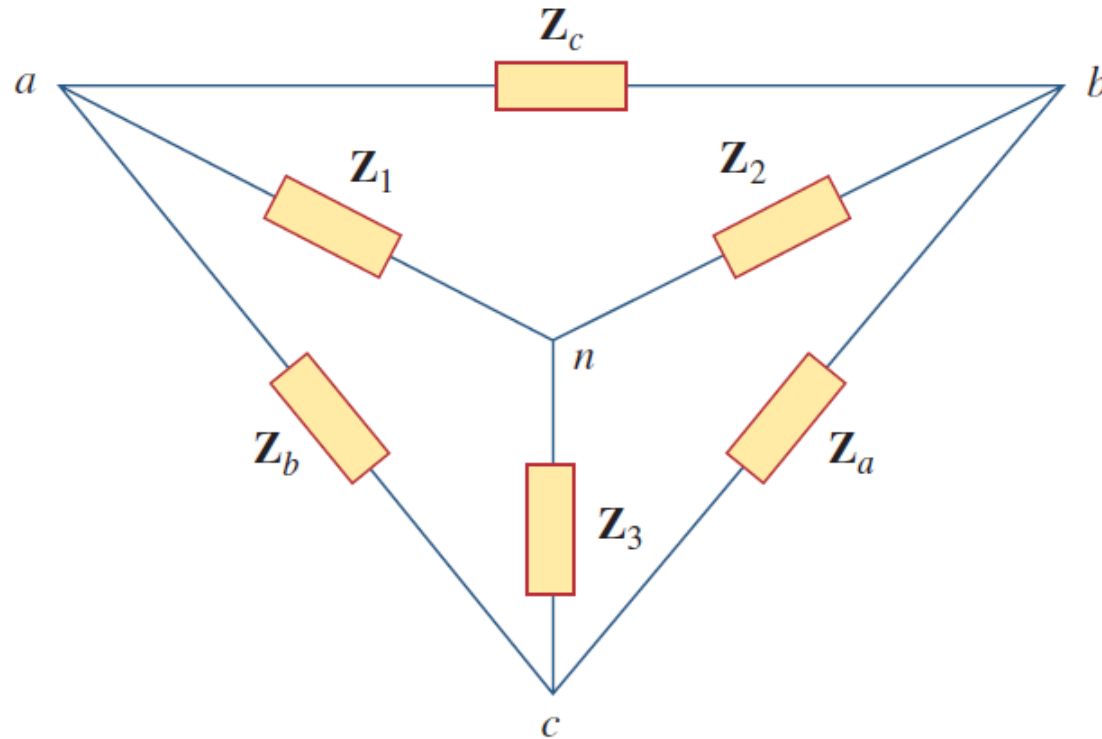
$$I = I_1 + I_2 + \cdots + I_N = V \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \cdots + \frac{1}{Z_N} \right)$$

$$\frac{1}{Z_{eq}} = \frac{I}{V} = \frac{1}{Z_1} + \frac{1}{Z_2} + \cdots + \frac{1}{Z_N}$$

$$Y_{eq} = Y_1 + Y_2 + \cdots + Y_N$$

Impedance Combinations

Y-Δ Conversion:



$$Z_a = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1}$$

$$Z_b = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2}$$

$$Z_c = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3}$$

Δ-Y Conversion:

$$Z_1 = \frac{Z_b Z_c}{Z_a + Z_b + Z_c}$$

$$Z_2 = \frac{Z_c Z_a}{Z_a + Z_b + Z_c}$$

$$Z_3 = \frac{Z_a Z_b}{Z_a + Z_b + Z_c}$$

Example: Impedance Combinations

- Q: find the input impedance in the circuit, assume that $\omega = 50 \text{ rad/s}$

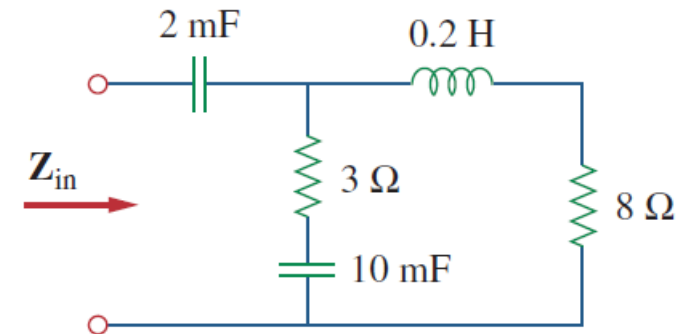
Solution:

Let

Z_1 = Impedance of the 2-mF capacitor

Z_2 = Impedance of the 3- Ω resistor in series with the 10-mF capacitor

Z_3 = Impedance of the 0.2-H inductor in series with the 8- Ω resistor



Then

$$Z_1 = \frac{1}{j\omega C} = \frac{1}{j50 \times 2 \times 10^{-3}} = -j10 \Omega$$

$$Z_2 = 3 + \frac{1}{j\omega C} = 3 + \frac{1}{j50 \times 10 \times 10^{-3}} = (3 - j2) \Omega$$

$$Z_3 = 8 + j\omega L = 8 + j50 \times 0.2 = (8 + j10) \Omega$$

The input impedance is

$$\begin{aligned} Z_{in} &= Z_1 + Z_2 \parallel Z_3 = -j10 + \frac{(3 - j2)(8 + j10)}{11 + j8} \\ &= -j10 + \frac{(44 + j14)(11 - j8)}{11^2 + 8^2} = -j10 + 3.22 - j1.07 \Omega \end{aligned}$$

Thus,

$$Z_{in} = 3.22 - j11.07 \Omega$$

Summary

- A sinusoid is a signal in the form of the sine or cosine function, i.e., $v_t = V_m \cos(\omega t + \phi)$
- A phasor is a complex quantity that represents both the amplitude and the phase of a sinusoid (missing out the angular frequency), denoted with boldface letter: $\mathbf{V} = V_m \angle \phi$
- In an ac circuit, the phase of voltage relative to the current is always fixed: in phase for resistor; leading ahead 90 degree for inductor and lagging behind 90 degree for capacitor
- Generalized Ohm's law holds for all three type of passive elements
- Kirchhoff's laws are still valid for phasors
- The techniques of voltage/current division, series/parallel combination of impedance/admittance, circuit reduction, and wye-delta transform all apply to ac circuit analysis